

Alisha Walunj

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SUMMARY

Software Engineer with 4 years of experience and an M.S. in Computer Science from Purdue University, specializing in GenAI, RAG systems, and scalable backend development. Experienced in building production AI applications using Python, Java, FastAPI, Spring Boot, LangChain, and Node.js across Azure and AWS. Skilled in cloud-native APIs, vector search, LLM orchestration and scalable REST, GraphQL, SOAP and microservices architectures.

EDUCATION

Purdue University | USA — Master of Science in Computer Science (M.S)

Aug 2023 - May 2025

Savitribai Phule Pune University | India — Bachelor of Engineering in Information Technology (B.E.)

Aug 2016 - May 2020

EXPERIENCE

One Community

USA

Software Engineer

Jun 2025 – Present

- Designed and deployed a production-grade multi-modal GenAI RAG solution on Azure using Python, LangChain and FastAPI, leading end-to-end delivery from stakeholder requirement gathering to scalable live API deployment.
- Integrated BGE embeddings with Pinecone and Weaviate vector databases, applying hybrid search and re-ranking strategies to enhance semantic retrieval of housing plans, property features and resident information, improving query relevance and retrieval productivity by 75%.
- Built an A/B testing framework benchmarking precision@k and MRR across embedding configurations, cutting p95 latency by 40% through data-driven prompt and retrieval optimization.
- Deployed a production-grade multi-modal AI application for volunteer management using MCP-based tool orchestration and intelligent LLM routing, ensuring high reliability, low-latency responses and seamless user interactions through canary deployments.

Capgemini

India

Associate Consultant

Jan 2023 – Aug 2023

- Improved SQL performance using stored procedures and foreign-key indexing, reducing DB round-trips and primary database load by 70%.
- Collaborated with QA to create test cases/documentation, conducted peer code reviews, and partnered with MetLife stakeholders to streamline Agile delivery.
- Configured Azure DevOps pipelines to build Docker images, push to Azure Container Registry and trigger deployments on every code merge.
- Extended a Spring Boot (REST) application to support legacy SOAP integrations by configuring and consuming third-party web services via JAX-WS over HTTP.
- Developed unit test suites using JUnit and Mockito achieving 90% code coverage, and integrated SonarQube quality gates into Azure CI/CD pipelines to automatically enforce code quality standards across all deployments.

Senior Software Engineer

Jan 2022 – Dec 2022

- Designed and developed REST APIs in Java Spring Boot for financial services, handling 1M+ daily requests secured with JWT and OAuth authentication, improving system efficiency by 35%.
- Built a RabbitMQ producer-consumer pipeline in MetLife's Java J2EE backend to asynchronously trigger transactional emails (payment confirmations, statements, balance alerts), decoupling the notification service from the core financial flow.
- Refactored legacy Java codebase across 8 admin systems, eliminating 1200+ lines of redundant code using OOP principles, lambdas and optimized data structures & algorithms to handle 10k+ daily transactions.

Software Engineer

Nov 2020 – Dec 2021

- Implemented shell scripts using Unix to automate ETL pipelines to extract, clean and load data into Oracle and MySQL database across multiple admin systems.
- Resolved N+1 query issues in a high-traffic Java backend using JPA and native SQL.
- Configured distributed logging across 5 microservices using Log4J, enabling per-function latency monitoring and improved observability.

PROJECTS

TaskOps | Spring Boot, Next.js, GraphQL, Apollo Federation, OAuth, PostgreSQL, RabbitMQ | Github Link

May 2025

- Designed a federated GraphQL architecture using Apollo Federation to unify multiple services, modeling relational data in PostgreSQL and securing APIs with OAuth2 + JWT.

Mastodon TutorAI | Python, MongoDB, ChromaDB, LLAMA, OpenAI API, Streamlit, Docker | Github Link

Dec 2024

- Built a RAG-based system using OpenAI APIs and vector embeddings (ChromaDB) to generate context-aware responses and quizzes from uploaded documents.